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Patrick Sicurello

EDUCATION

University of California Berkeley Bachelor of Arts in Physics and Applied Mathematics	Berkeley, CA 08/2016 – 12/2020
University of California San Diego Certificate in Machine Learning	Remote 09/2021 – 03/2022
Johns Hopkins University Master of Applied Mathematics	Baltimore, MD 08/2023 – Present

TECHNICAL SKILLSET

Languages/Frameworks: Python, PyTorch, PyTorch Lightning, TensorFlow, HuggingFace

Scientific ML: Diffusion Models (DDPM), Physics-Informed Neural Networks (PINNs), Neural Operators, Geospatial ML, Surrogate Modeling, Algorithm Discovery

Data & Systems: PySpark, Ibis, Kedro, MLflow, Weights & Biases, AWS S3, Docker, Kubernetes

Other: Bayesian Optimization, GIS (QGIS), SQL, Java

PROFESSIONAL EXPERIENCE

8/2022-Present Johns Hopkins University Applied Physics Lab Laurel, MD

Machine Learning Researcher/Engineer

- Developed multimodal deep learning models for global road activity using GIS and OSM data, allowing us to model greenhouse gas (GHG) emissions at a global scale as part of Climate TRACE coalition work
- Developed global-scale inference pipeline for road transportation emissions model using Kedro and Ibis, outputting ~4TB data product
- Designed and implemented novel multi-headed physics-informed neural network (PINN) to predict sea ice drift vectors in dynamic environments
- Developed a complex-valued diffusion model (DDPM) for SAR remote sensing imagery to preserve physically meaningful phase structure lost in standard real-value diffusion models
- Fine-tuned geospatial foundation models (GeoFM) for GIS image segmentation

- and classification, for downstream task of curating a global inventory of wastewater treatment ponds
- Performed uncertainty quantification for calibration of additive manufacturing processes, allowing for laser correction in regions of the build plate that resulted in defects
- Performed independent verification and validation (IV&V) for DARPA DIAL program, evaluating performer methods in their ability to accelerate scientific discovery through newly discovered numerical methods
- Designed and implemented graph-based models (GraphSAGE and Conditional Markov Network) for predicting spotted lanternfly populations in the Northeast US based on climate and co-occurrence of plant species
- Routinely wrote grant proposals and pitched to potential sponsors for research funding through quarterly presentations (e.g., NASA Cryosphere, NOAA, NRO)
- Work presented at American Meteorological Society Conference, AGU, and COP29
- Mentored high school and undergraduate interns on developing an LLM-based software for editing structured space-sector data

10/2021 – 7/2022

HedgehogAI

Palo Alto, CA

Machine Learning Engineer

- Trained and deployed industry NLP model for named-entity recognition on pet insurance claim documents
- Support client teams with model implementation and assisting them in troubleshooting deployment issues
- Serve as primary point of contact between HedgehogAI and clients

10/2020 – 11/2021

Independent

Berkeley, CA

Freelance Developer

- Developed Jupyter Notebook plugins for remote clients to improve onboarding and work pipelines
- Wrote Data Science homework assignments and autograder for the University of Maryland Global Campus
- Performed data cleaning and pre-processing for clients
- Developed SQL and NoSQL databases for managing independent card-trading store inventory and purchases